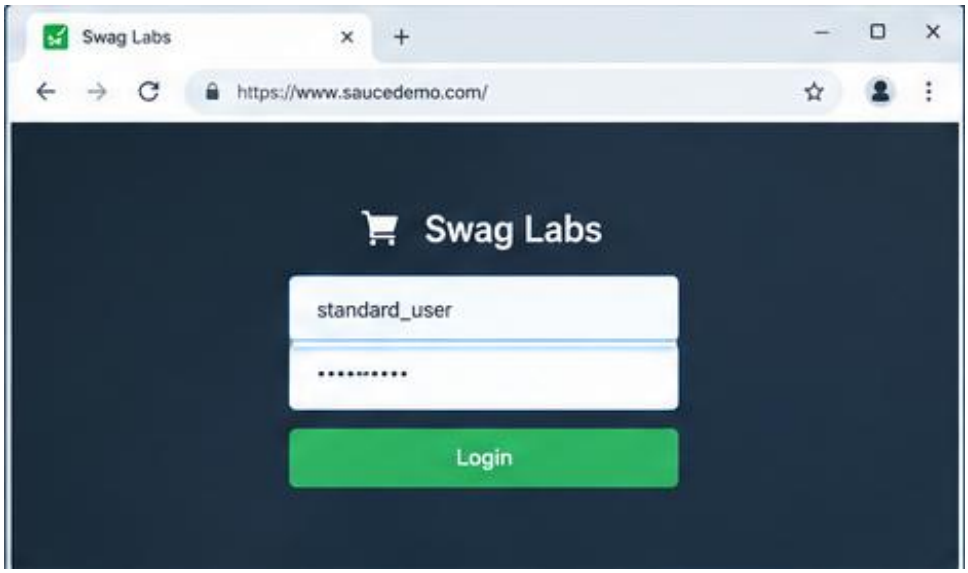
Module Name: Authentication / Login - Automated Login Testing using Python Selenium
Test Case ID: st_15
Tester Name: Aman Sevak
Webpage URL: https://www.saucedemo.com/
Tools Used: Python, Selenium WebDriver, PyCharm, Google Chrome

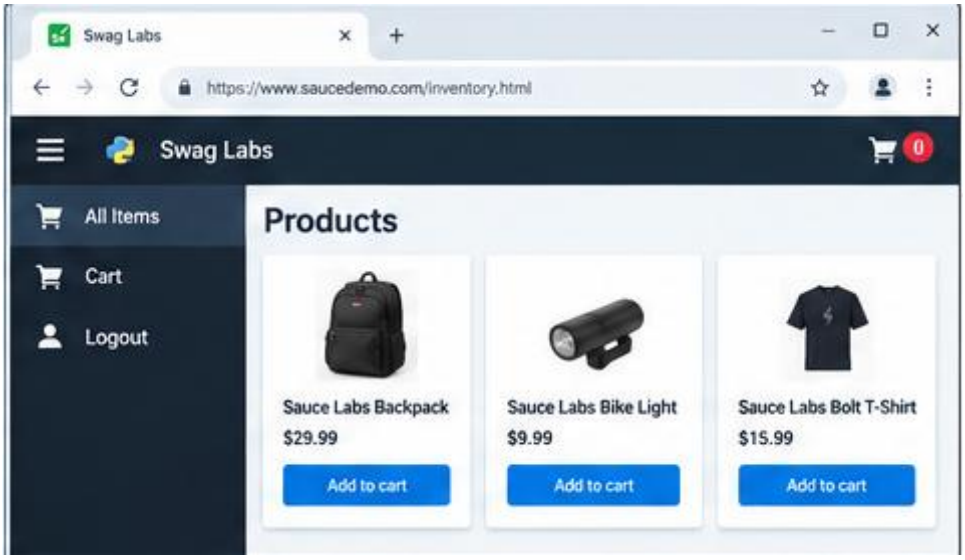
- ## Scenario ID: 1a

Scenario: Valid login (TC-01)
Test Steps:

1. Launch Google Chrome and open https://www.saucedemo.com/
2. Enter Username: standard_user
3. Enter Password: secret_sauce
4. Click on Login
5. Validate that the URL contains "inventory" and the page title is "Products"

Expected / Actual Result: User logged in successfully and Products (inventory) page opened.
Status: **Pass**
Screenshots:



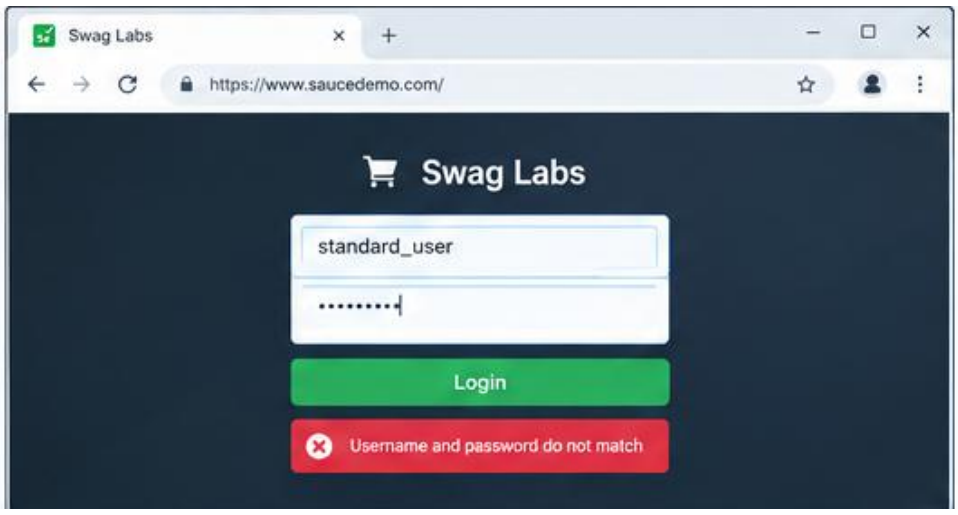


- **Scenario ID: 1b**

Scenario: Login with invalid password (TC-02)
Test Steps:

1. Open the login page
2. Enter Username: standard_user
3. Enter Password: wrong_password
4. Click on Login
5. Verify the error message is displayed

Expected / Actual Result: Error message "Username and password do not match" displayed.
Status: **Pass**
Screenshots:

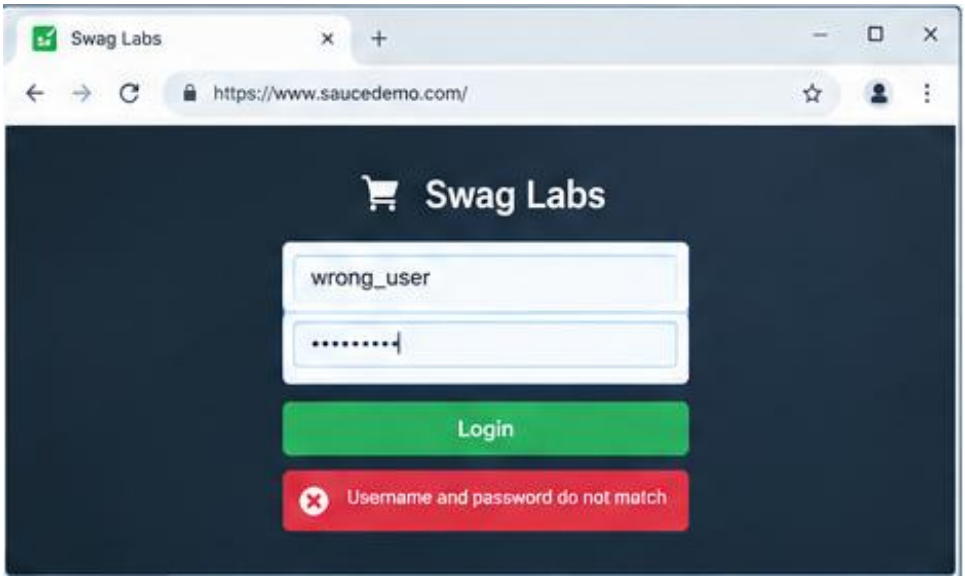


- **Scenario ID: 1c**

Scenario: Login with invalid username (TC-03)
Test Steps:

1. Open the login page
2. Enter Username: wrong_user
3. Enter Password: secret_sauce
4. Click on Login
5. Verify the error message is displayed

Expected / Actual Result: Error message "Username and password do not match" displayed.
Status: **Pass**
Screenshots:

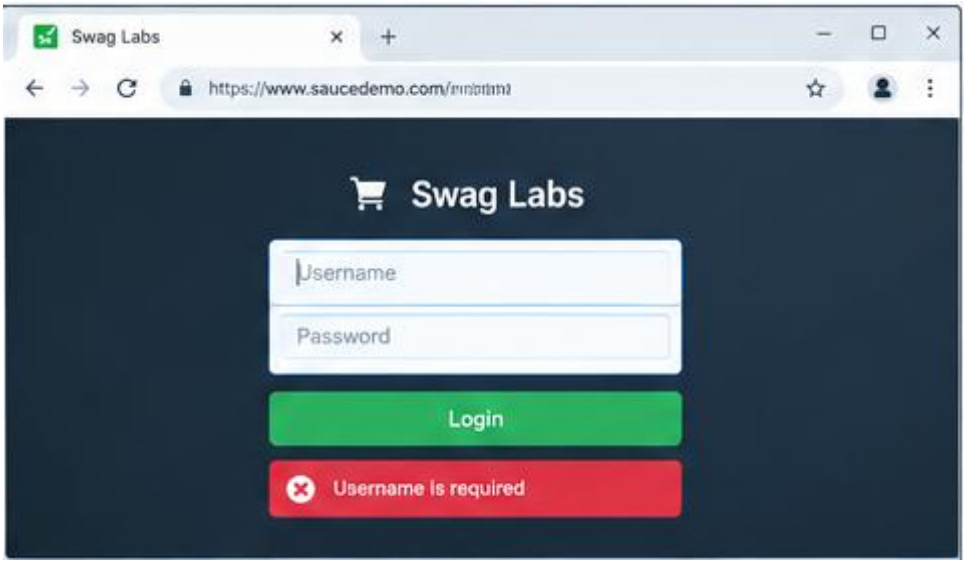


- **Scenario ID: 1d**

Scenario: Login with empty credentials (TC-04)
Test Steps:

1. Open the login page
2. Leave Username and Password blank
3. Click on Login
4. Verify the validation message is displayed

Expected / Actual Result: Validation message "Username is required" displayed.
Status: **Pass**
Screenshots:

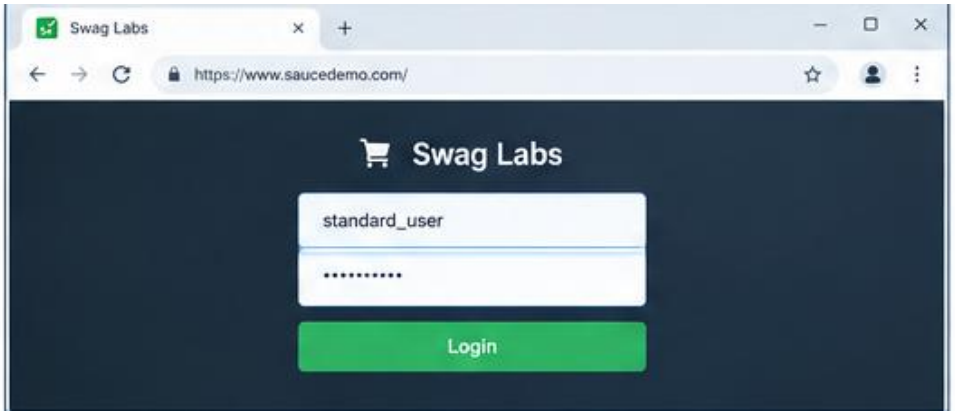


- **Scenario ID: 1e**

Scenario: Logout (TC-05)
Test Steps:

1. Login with valid credentials
2. Click on the menu (burger) button
3. Click on Logout
4. Verify the login page is displayed

Expected / Actual Result: User logged out and returned to the login page.
Status: **Pass**
Screenshots:



- **Source Code (tutorial15.py)**

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)

URL = "https://www.saucedemo.com/"
VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"



def start_browser():
    driver = webdriver.Chrome()
    driver.maximize_window()
    return driver



def login(driver, username, password):
    driver.get(URL)
    wait = WebDriverWait(driver, 10)
    username_box = wait.until(EC.visibility_of_element_located((By.ID, "user-
name")))
    password_box = wait.until(EC.visibility_of_element_located((By.ID,
"password")))
    login_button = wait.until(EC.element_to_be_clickable((By.ID, "login-button")))
    username_box.clear()
    username_box.send_keys(username)
    password_box.clear()
    password_box.send_keys(password)
    login_button.click()



def get_error(driver):
    return WebDriverWait(driver, 10).until(
        EC.visibility_of_element_located((By.CSS_SELECTOR, "h3[data-
test='error']"))
    )



# TC-01 Valid Login
def test_valid_login():
    driver = start_browser()
    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)
        WebDriverWait(driver, 10).until(EC.url_contains("inventory"))
        assert "inventory" in driver.current_url
        title = WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located((By.CLASS_NAME, "title"))
        )
        assert title.text == "Products"
        print("TC-01 Valid Login: PASS")
        print("Authentication Assertion: PASS")
    except AssertionError:
        print("TC-01 Valid Login: FAIL")
    except NoSuchElementException:
        print("NoSuchElementException handled")
    except TimeoutException:
        print("TimeoutException handled")
    except ElementNotInteractableException:
```

```
        print("ElementNotInteractableException handled")
    except StaleElementReferenceException:
        print("StaleElementReferenceException handled")
    finally:
        driver.quit()


# TC-02 Invalid Password
def test_invalid_password():
    driver = start_browser()
    try:
        login(driver, VALID_USERNAME, "wrong_password")
        assert "Username and password do not match" in get_error(driver).text
        print("TC-02 Invalid Password: PASS")
    except AssertionError:
        print("TC-02 Invalid Password: FAIL")
    except TimeoutException:
        print("Error message was not displayed")
    finally:
        driver.quit()


# TC-03 Invalid Username
def test_invalid_username():
    driver = start_browser()
    try:
        login(driver, "wrong_user", VALID_PASSWORD)
        assert "Username and password do not match" in get_error(driver).text
        print("TC-03 Invalid Username: PASS")
    except AssertionError:
        print("TC-03 Invalid Username: FAIL")
    except TimeoutException:
        print("Error message was not displayed")
    finally:
        driver.quit()


# TC-04 Empty Credentials
def test_empty_credentials():
    driver = start_browser()
    try:
        driver.get(URL)
        WebDriverWait(driver, 10).until(
            EC.element_to_be_clickable((By.ID, "login-button"))
        ).click()
        assert "Username is required" in get_error(driver).text
        print("TC-04 Empty Credentials: PASS")
    except AssertionError:
        print("TC-04 Empty Credentials: FAIL")
    except TimeoutException:
        print("Validation message was not displayed")
    finally:
        driver.quit()


# TC-05 Logout
def test_logout():
    driver = start_browser()
    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)
        WebDriverWait(driver, 10).until(EC.url_contains("inventory"))
        WebDriverWait(driver, 10).until(
            EC.element_to_be_clickable((By.ID, "react-burger-menu-btn"))
        ).click()
        WebDriverWait(driver, 10).until(
            EC.element_to_be_clickable((By.ID, "logout_sidebar_link"))
```

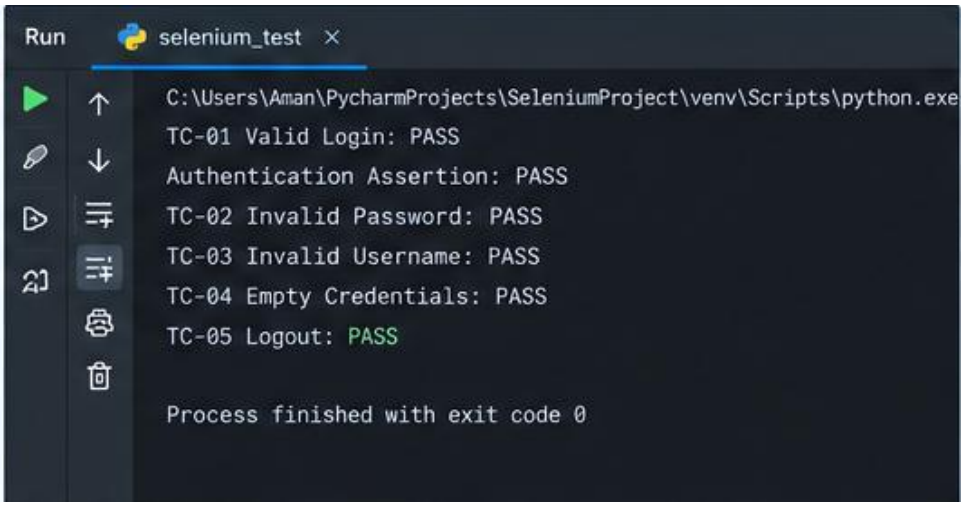
```
        ).click()
        WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located((By.ID, "login-button"))
        )
        assert "saucedemo.com" in driver.current_url
        print("TC-05 Logout: PASS")
    except AssertionError:
        print("TC-05 Logout: FAIL")
    except TimeoutException:
        print("Logout operation timed out")
    finally:
        driver.quit()


# Execute all test cases
test_valid_login()
test_invalid_password()
test_invalid_username()
test_empty_credentials()
test_logout()
```

- **Console Output (PyCharm)**

```
TC-01 Valid Login: PASS
Authentication Assertion: PASS
TC-02 Invalid Password: PASS
TC-03 Invalid Username: PASS
TC-04 Empty Credentials: PASS
TC-05 Logout: PASS

Process finished with exit code 0
```



- **Summary Of Test Results:**

Total Test Cases- 5
Passed- 5
Failed- 0
Pass %- 100%

Conclusion and Recommendations:

Pass Rate: 100%, All Tests Passed

The Selenium WebDriver script successfully automated valid login, invalid password, invalid username, empty credentials and logout scenarios of the SauceDemo application. All

assertions passed, error messages were validated correctly and exceptions were handled without abrupt termination of the script.

Issues Found

No issues or defects were found during execution of the automated test scenarios.

Recommendations

1. **Add More Negative Test Cases**
   Extend the script with locked_out_user login, special-character input and password-only / username-only empty field scenarios.
2. **Use Page Object Model**
   Separate locators and actions into page classes to make the script easier to maintain.
3. **Run as Regression Suite**
   Re-execute the script after every change to confirm that existing authentication functionality is unaffected.

- **Bug Reports:**

No bugs were reported for this practical, as all test cases passed.